Letter of Recommendation for Xuan Thanh Trinh

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July 22, 2026

To: Ph.D. Selection Committee
Doctoral Program in Electrical, Electronics and Computer Engineering
College of Engineering and Computer Science (CECS)
University of Michigan–Dearborn, MI 48128, USA

Dear Admissions Committee,

I am Dr. Tran Thanh Son, a Lecturer in the School of Electrical and Electronic Engineering at Hanoi University of Science and Technology (HUST). Having earned my Ph.D. from the Shibaura Institute of Technology in Japan, my research focuses heavily on power system optimization and microgrid architectures. I served as Xuan Thanh Trinh's direct research mentor on a complex project regarding peer-to-peer energy trading frameworks. Guiding him through the rigorous process of developing this research into a full academic manuscript provided me with profound insights into his technical and intellectual caliber. Based on this exceptional performance, I write to you today to offer my highest recommendation for his admission to the Ph.D. program at the University of Michigan-Dearborn.

Thanh demonstrates a rare capacity for mathematical abstraction and research autonomy. Rather than passively applying formulas, he established the global convergence proof for an ADMM framework combined with Second-Order Cone Programming (SOCP) to accommodate non-linear grid constraints. Furthermore, he consistently masters advanced theoretical frameworks outside the standard undergraduate curriculum with minimal supervision. This mathematical intuition, coupled with his ability to navigate dense academic literature, proves he is fully equipped for the rigor of Ph.D. research.

Furthermore, Thanh exhibits exceptional system-level thinking. He designed evaluation scenarios to analyze the physical coupling between microgrids and the utility network under high renewable penetrations. To validate these models, he stress-tested his algorithms on the IEEE 123-bus system, successfully resolving critical ADMM non-convergence issues via dynamic penalty mechanisms. His ability to anchor abstract designs with realistic grid constraints provided the empirical foundation for our manuscript, currently under review at the journal Sustainable Energy, Grids and Networks (SEGAN).

Ultimately, what sets Thanh apart is his sustained methodological resilience during complex simulation cycles. When debugging large-scale grid networks, he constructively treats algorithmic non-convergence as diagnostic data rather than failures. Furthermore, even under tight publication deadlines, he systematically traces numerical instabilities to their theoretical root causes instead of relying on unverified heuristics. This analytical endurance proves he is exceptionally well-equipped for the rigor of a Ph.D. program.

In summary, Thanh uniquely combines mathematical rigor, system-level thinking, and methodological resilience. Having worked closely with him as a co-author, I give him my highest recommendation for admission to your Ph.D. program without reservation. I am confident he will be an immediate and highly productive contributor to your research group. Please feel free to contact me if you require any further information.

Sincerely,

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